

Doctoral thesis report on the thesis
Threat Detection in Network Traffic using Time Series Analysis
by doctoral candidate Josef Koumar

The thesis has been submitted to the Faculty of Information Technology, Czech Technical University in Prague.

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Timeliness and relevance

This dissertation is both timely and highly relevant. The adoption of encryption in modern network traffic has rendered traditional DPI methods ineffective, creating a need for alternative detection approaches. By leveraging time series analysis of network metadata, the thesis develops detection mechanisms that operate without access to plaintext content, making its contributions directly applicable to today's high-speed networks. The experimental evaluation on cryptomining and botnet command-and-control detection is particularly valuable, as these represent growing activities in today's networks.

Structure and presentation

The thesis is well organized across seven chapters. Chapter 1 introduces the topic and sets out the research goals, while Chapter 2 provides thorough background on time series analysis and network traffic monitoring, along with a comprehensive discussion of the related state of the art. Chapter 3 defines the four types of network traffic time series considered throughout the work, laying a clear conceptual foundation for the chapters that follow. Chapter 4 presents and evaluates two sets of time series features against publicly available datasets, with a commendable division between the "ideal" feature set (SPLT) and the more resource-conscious, deployment-oriented features (NetTiSA). The description of the individual features could have been more detailed, though this does not significantly detract from the overall contribution.

Chapter 5 proposes a periodicity-based approach to traffic analysis and applies it to classification and identification tasks, including two extensive and relevant use cases: cryptomining and botnet traffic classification. Each use case is systematically treated with a clear presentation of the methodology, evaluation setup, and results. Evaluations include the impact of time window size and detection threshold, and the botnet detection results are compared against existing methods on the CTU-13 dataset.

Chapter 6 opens with the presentation of a new dataset constructed using a technique that aggregates time series for inbound and outbound traffic from the CESNET3 network. The anomalies identified in this dataset are then described, followed by network traffic forecasting using SARIMA. Building on this, a novel modular Network Outlier Detection System (NODS) is proposed and evaluated on the dataset. NODS is implemented as NEMEA modules, trained on historical data to predict future traffic behavior, and detections are made by comparing these predictions against actual observations. NODS offers three detection model implementations: Mean Outlier Detection, Differential Outlier Detection, and SARIMA. Experimental evaluation shows that most outliers are detected based on the number of flows per aggregation interval. To reduce the volume of outliers requiring investigation, the author proposes grouping them into incidents.

In the second part of chapter 6, the use of LLMs to explain detected anomalies is proposed. The LLM output is assessed against annotations provided by human experts. Experiments demonstrate that even smaller LLMs can deliver competent results when deployed as anomaly analyst agents, and that they can meaningfully assist in real-time alert triaging. Compared to the other contributions, the work on LLMs gives the impression to be more exploratory.

Overall, the quality of presentation is good, with clear explanations supported by well-chosen figures and diagrams. Two minor presentational issues should be mentioned here: a number of bibliography entries contain incomplete information, missing venue or year of publication, and the absence of numbered subsections makes cross-referencing cumbersome. Neither issue is critical.

Dissemination

The dissemination effort accompanying this thesis is commendable. The candidate has published nine peer-reviewed articles directly relevant to the thesis [A.1–A.9], spanning both conference and journal venues. The conferences are well-established venues in the network management and measurement community, including CNSM, NOMS, and TMA, which are solid choices for this line of research. Encouragingly, the publications have already attracted a meaningful number of citations. Beyond the articles themselves, the candidate has made a substantial effort to release openly accessible datasets on Zenodo. Overall, the dissemination effort is positive and reflects a genuine commitment to open and reproducible research.

Methodology

The methodology underlying this thesis is sound. A particularly strong aspect is the focus on deployability: rather than proposing detection methods that work only under laboratory conditions, the candidate designs approaches that are mindful of real-world constraints such as bandwidth limitations and computational resources, as exemplified by the NetTiSA feature set and the implementation of NODS as NEMEA modules ready for integration into operational ISP infrastructure. The thesis makes extensive use of real network traffic datasets, including traffic captured from the CESNET production network, which lends the experimental results a credibility that lab-generated datasets alone cannot provide. The choice of use cases, especially in Chapter 5 (cryptomining and botnet C&C detection) is practically relevant. The experimental methodology pays attention to sensitivity analyses such as the impact of time window size and detection threshold and uses established benchmarks such as CTU-13 for comparative evaluation.

Contribution

The thesis makes several concrete contributions to the field of network threat detection. First, a taxonomy of network-derived time series is formalized. Building on this, time series analysis feature sets are proposed and shown to match or outperform related work across various binary and multiclass traffic classification tasks. A deployable, bandwidth-constrained variant of a feature set, called NetTiSA, is introduced for high-speed network environments. A novel periodicity-based detection approach is proposed and applied to multiple use cases, most notably achieving high precision across several datasets. On the anomaly detection side, the NODS system is proposed and deployed in a real ISP network. Finally, the use of LLMs for the automated explanation of detected anomalies is explored. Complementing these methodological contributions, the candidate has released several novel datasets.

Question for the defense

Have you looked at the long-term performance of your detection methods? Do you expect that retraining or adaptation of the detection parameters will be necessary after a certain time because the network traffic changes its characteristics?

Overall evaluation

This dissertation presents a coherent body of research that makes meaningful advances in network threat detection. The contributions are technically sound, practically oriented, and supported by experimental evaluation on real-world datasets.

The author of this dissertation proved the ability to conduct research and achieve scientific results. I therefore recommend the thesis for the presentation and defense with the aim of receiving the Ph.D. degree.

Louvain-la-Neuve, 10th May, 2026

A handwritten signature in black ink, consisting of a stylized 'R' followed by a period.

Ramin Sadre